

Mechanism Sessional

(ME29002)

DEPARTMENT OF MECHANICAL ENGINEERING
(3RD YEAR UG)



Indian Institute of Technology
KHARAGPUR

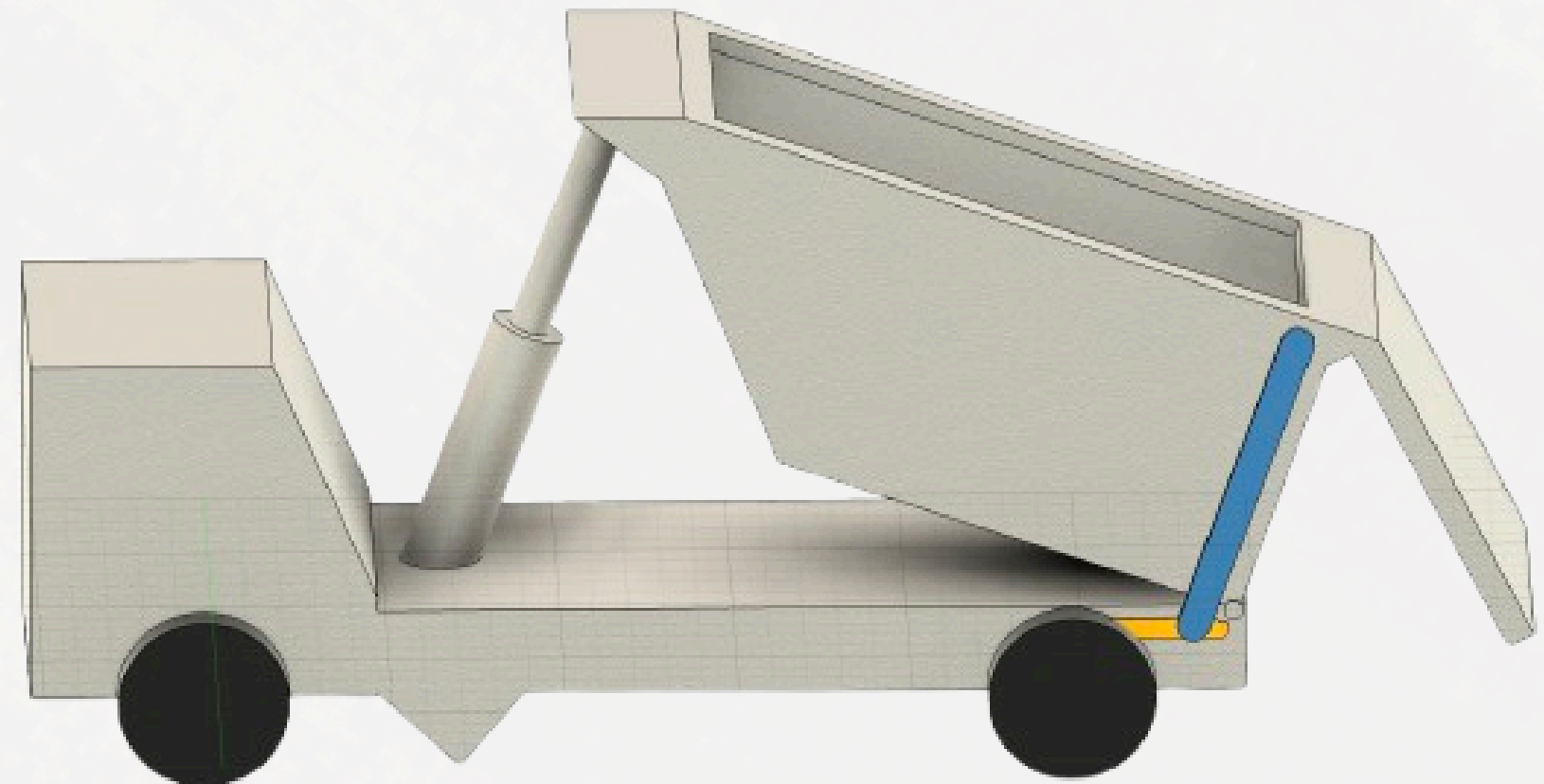
Topic :

Dumper Truck

Rear-Door Mechanism

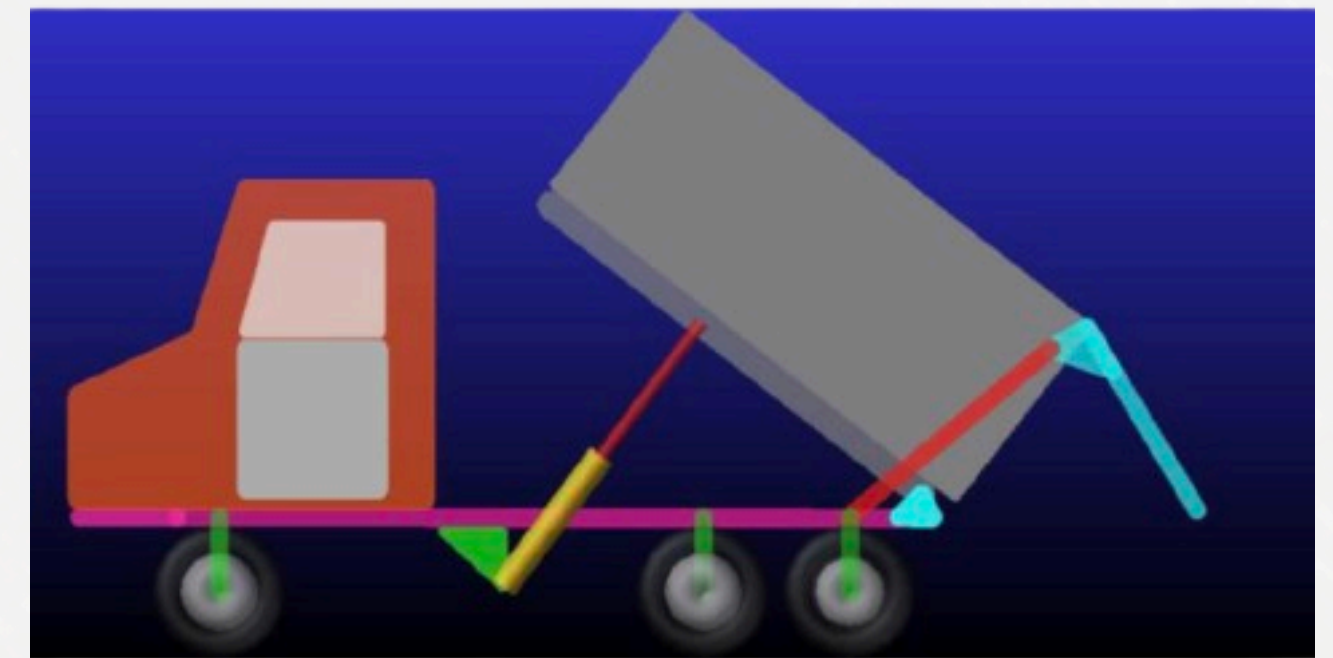
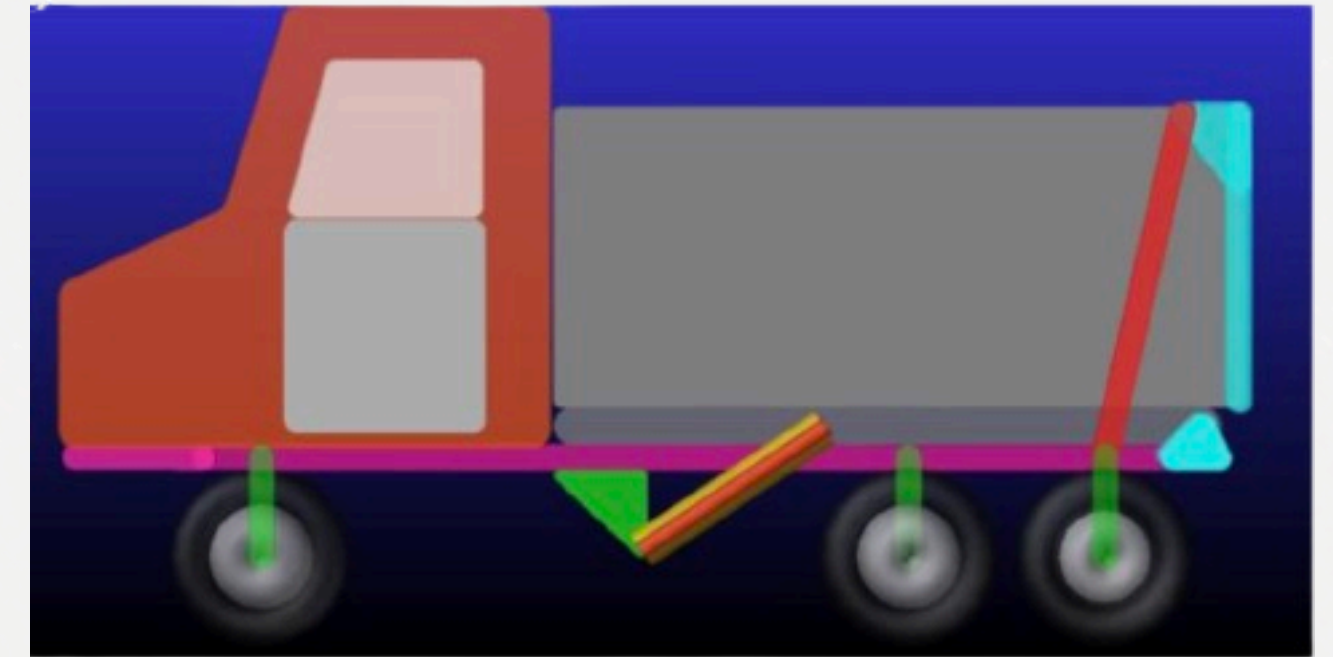
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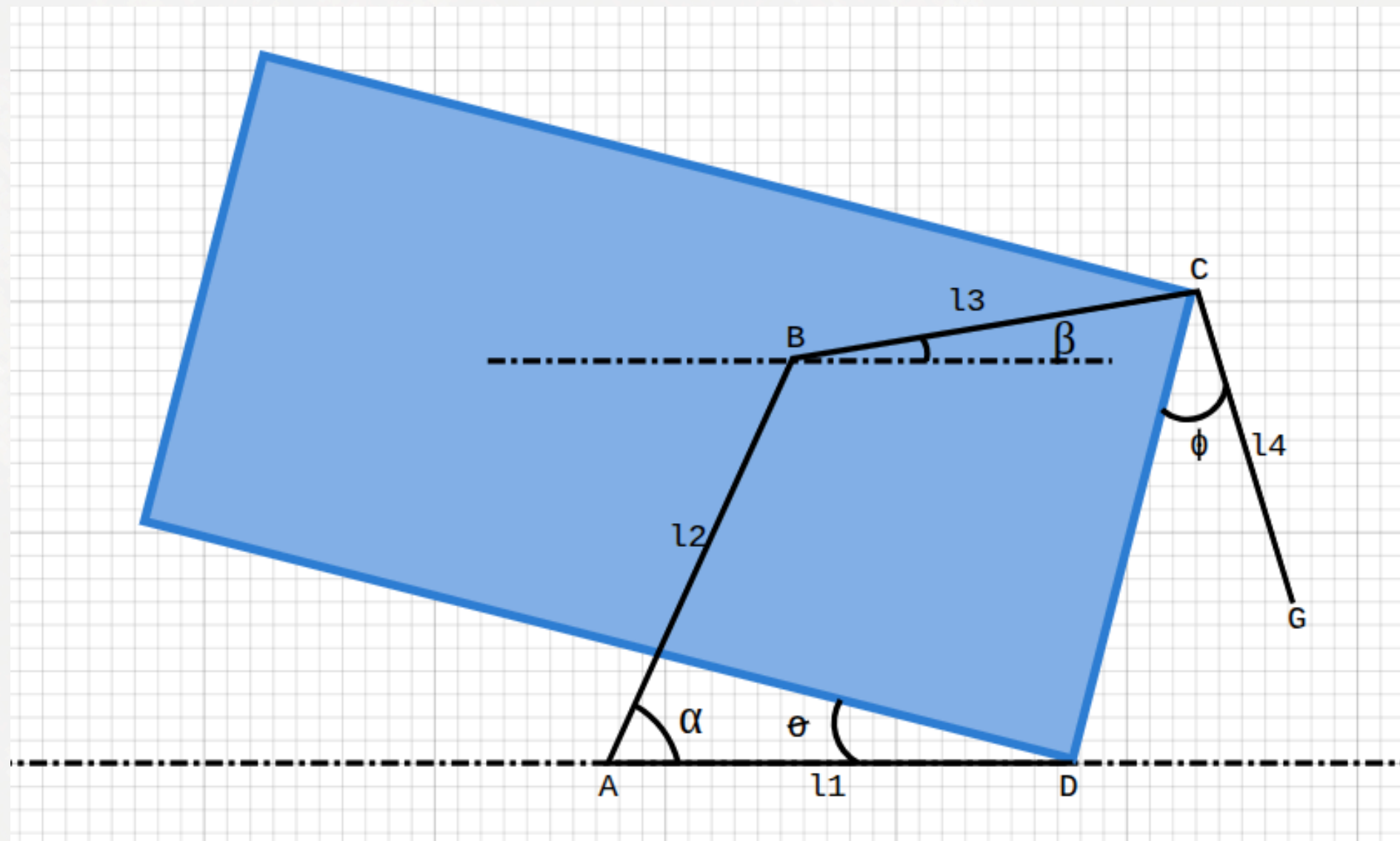
Introduction

Dumper trucks typically use manually unlocked, gravity-driven rear doors, which can be unsafe and inefficient during unloading. This project proposes a mechanized door system that opens automatically as the bin tilts, eliminating manual intervention. A simulator is developed to study the functional relationship between bin tilt angle and door opening angle, along with enhancements such as adjustable maximum opening, optimized linkage design, and an automatic lock–unlock feature for safer and smoother operation.



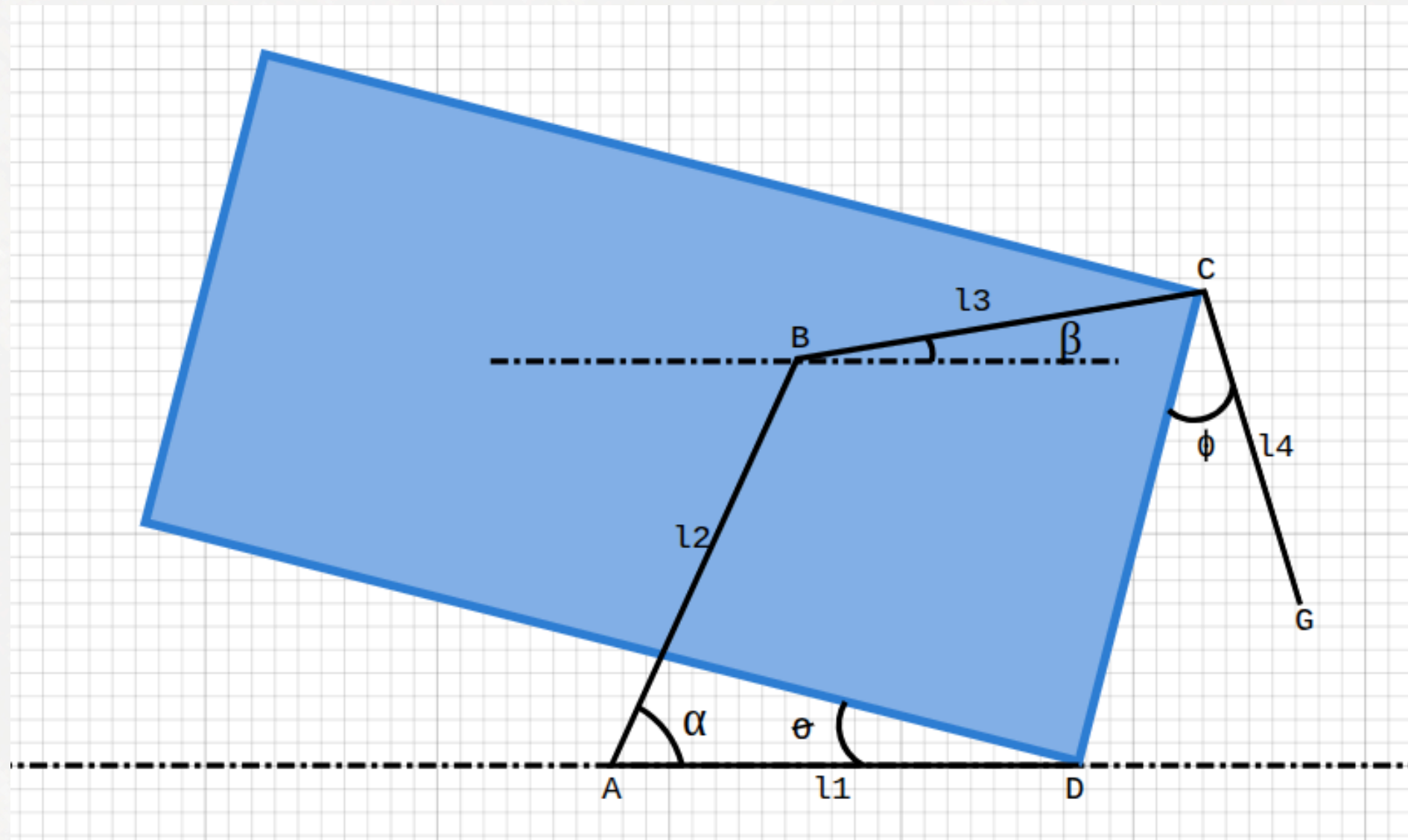
Synthesis of Mechanism

The dumper truck mechanism is illustrated through a simplified kinematic representation shown in the figure. This schematic emphasizes the rear-end door actuation system, which is driven by a dedicated actuator integrated into the lower chassis. The arrangement demonstrates the coordinated motion transfer and controlled kinematic response required for efficient and reliable door opening during tipping operations.



Parameters

- θ :The bin tilt angle
- Φ :The gate-open angle
- α :The angle between links L1 and L2
- β :The angle between link L3 and the horizontal



Relationship of angle α and β

$$a = \sin(90 - \theta)$$

$$b = \cos(90 - \theta) + \frac{l_1}{l_4}$$

$$c = \frac{l_1}{l_2} \cos(90 - \theta) + \frac{l_1^2 + l_2^2 + l_4^2 - l_3^2}{2l_2l_4}$$

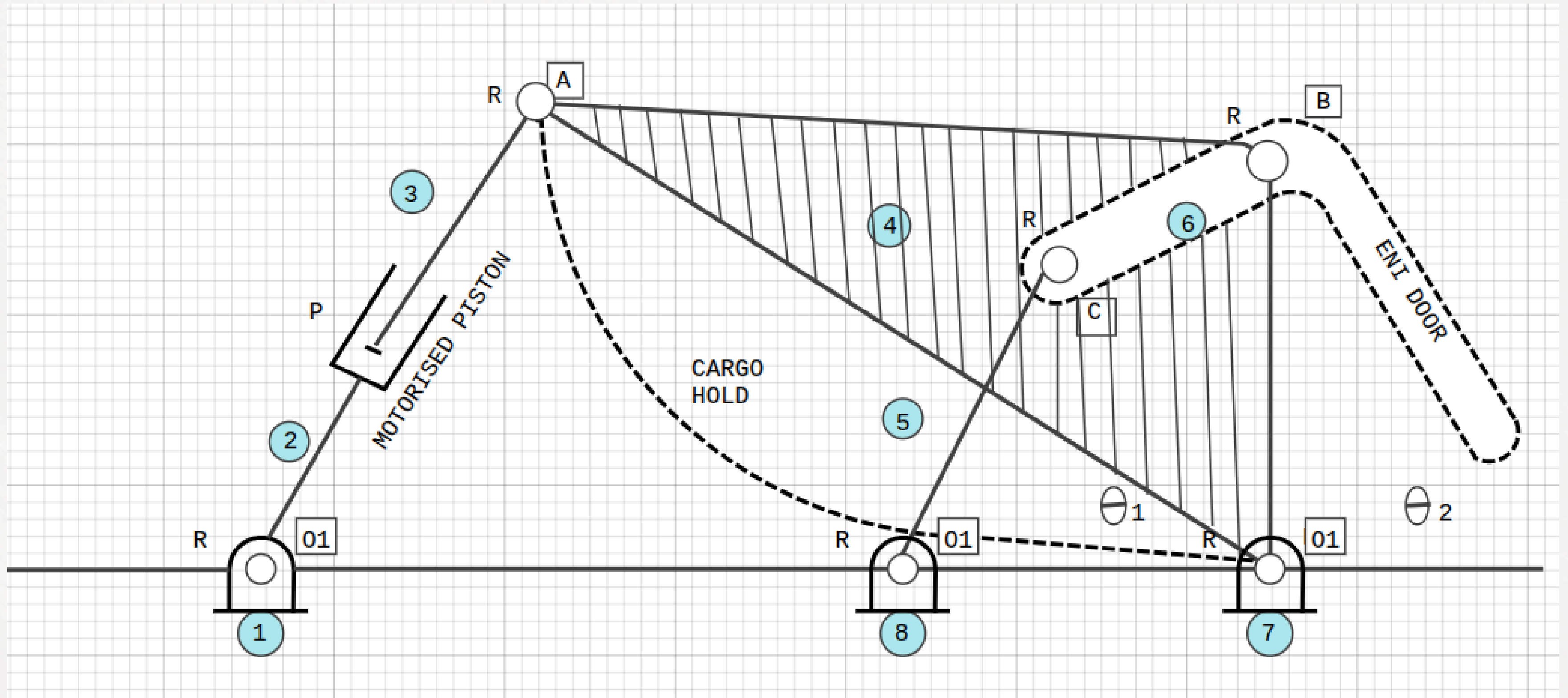
$$\alpha = 2 \tan^{-1} \left(\frac{a + \sqrt{a^2 + b^2 - c^2}}{b + c} \right)$$

$$\beta = \tan^{-1} \left(\frac{l_4 \sin(90 - \theta) - l_2 \sin \alpha}{l_1 + l_4 \cos(90 - \theta) - l_2 \cos \alpha} \right)$$

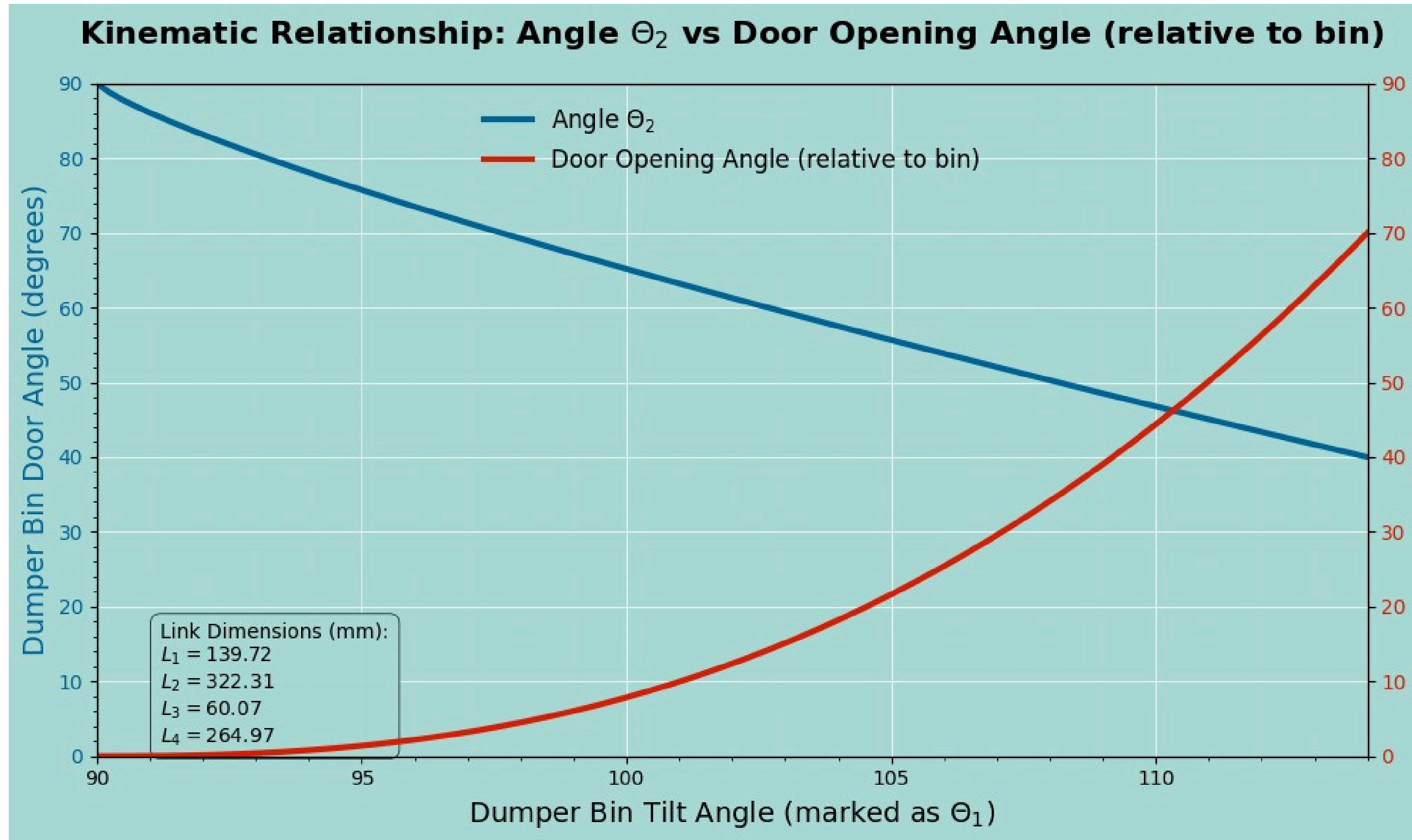
$$\phi = \theta + \beta$$

At the end we get the relation between bin tilt angle and gate angle.

Kinematic Diagram

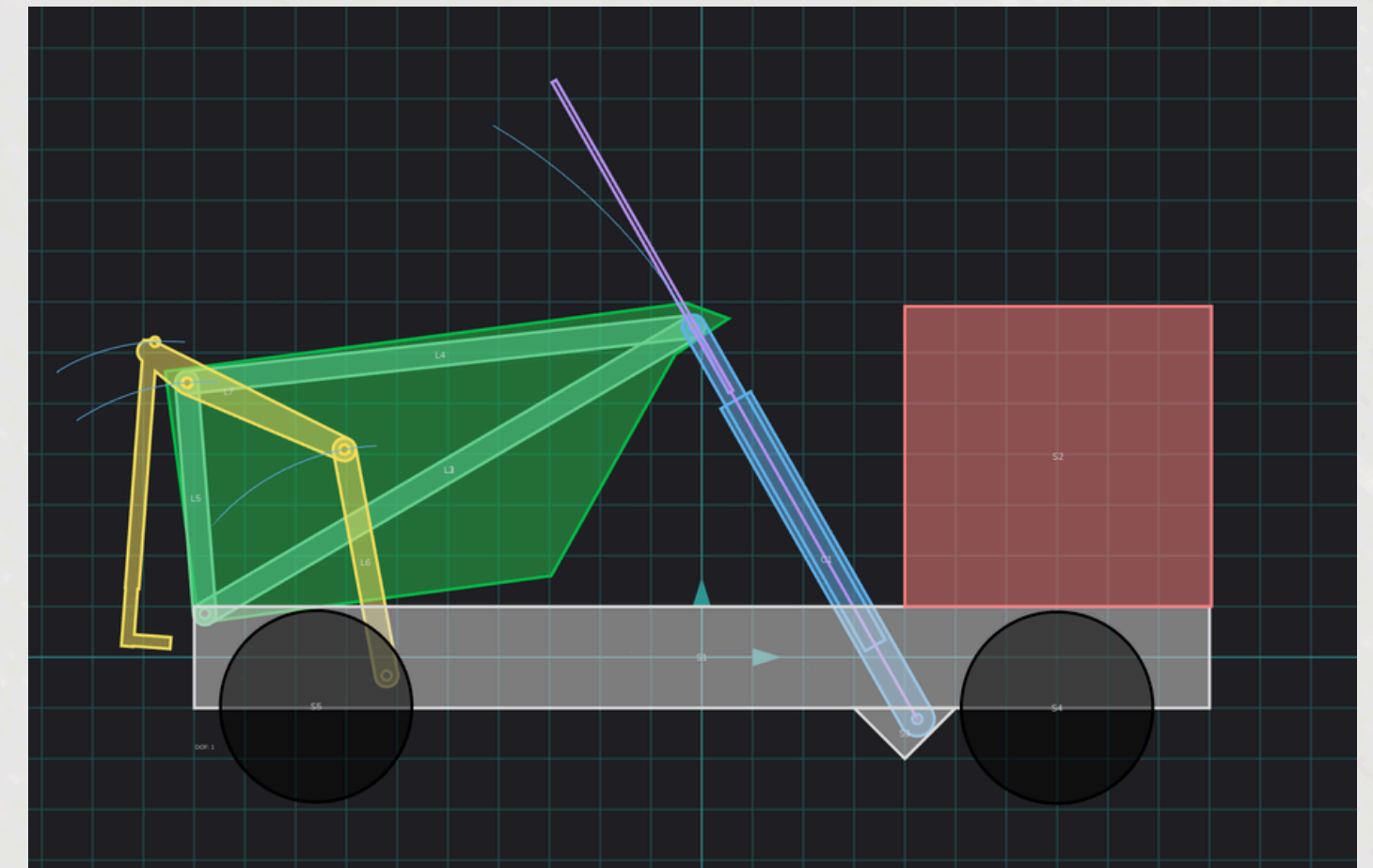
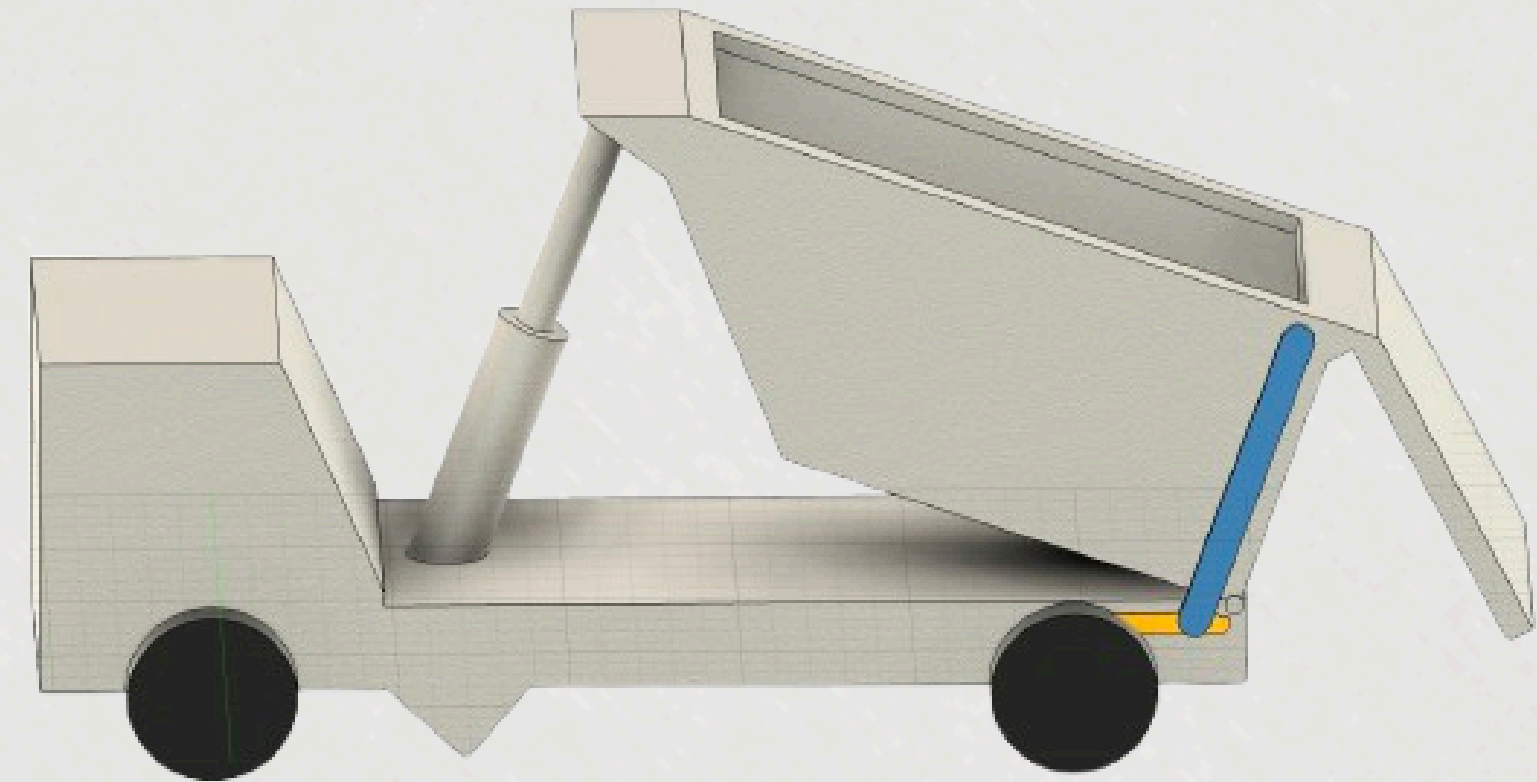


Graph



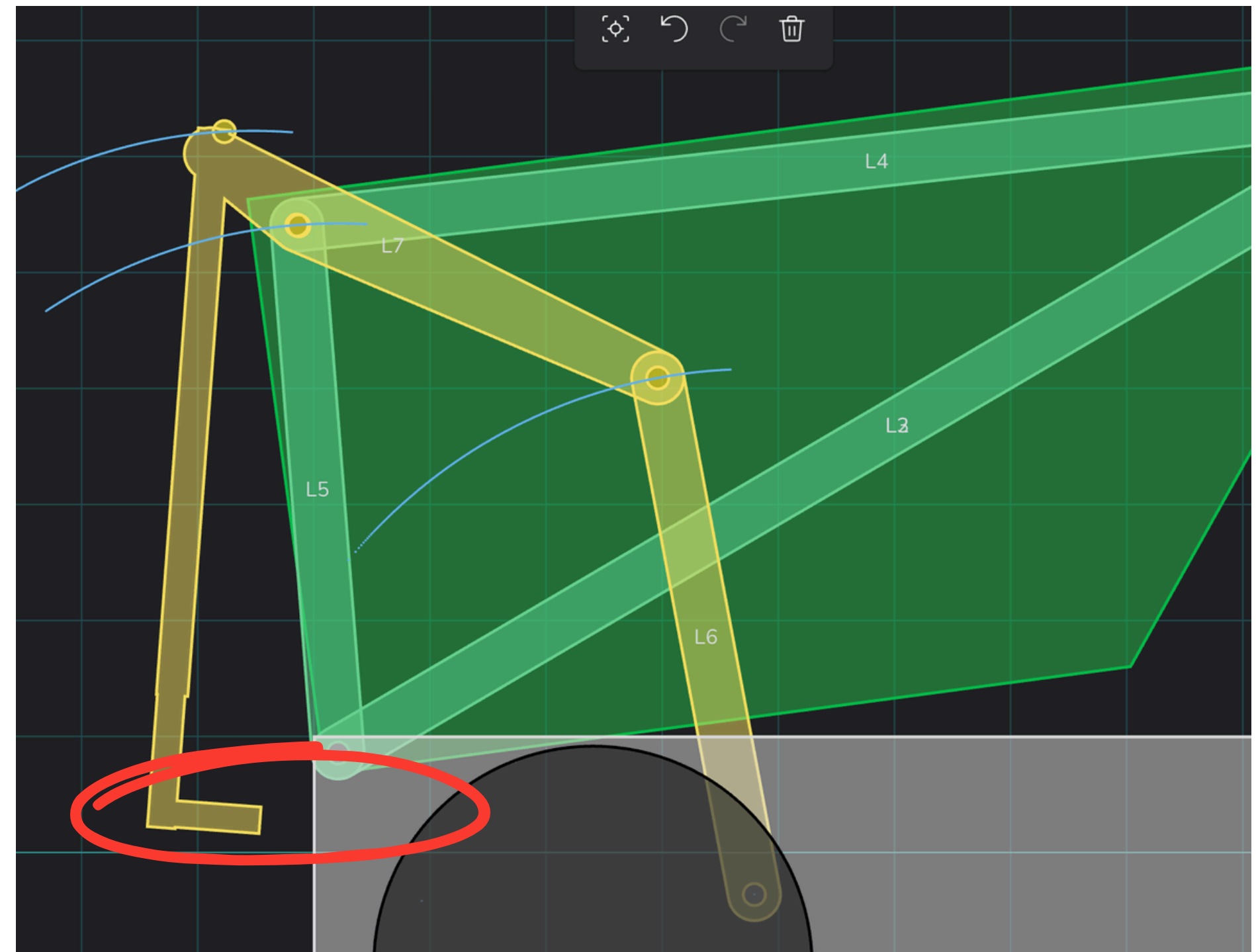
Simulation

- A full 2D linkage model was created to study the motion of the bin and tail-door.
- The hydraulic actuator is the single input (1 DOF) driving the entire system.
- As the actuator extends, the bin begins to tilt and simultaneously the lower link rotates, controlling the tail-door opening angle.
- Simulation verifies:
 - Smooth, interference-free motion
 - Proper sequencing (door opens only after bin starts lifting)
 - Full door clearance during dumping
 - Correct mapping between actuator stroke and bin/door angles



Innovation

- We introduce a simple mechanical lock that engages automatically when the garbage door returns to the 0° (closed) position.
- A small rectangular extension is added to the door. When the door closes, this extension drops into a slightly oversized slot provided in the truck base, securely locking the door.
- When the door begins to open, the extension is cam-lifted out of the slot by the natural motion of the mechanism—so the lock releases automatically without any actuator or manual effort.
- A weak embedded magnet inside the slot and a metal pad on the door extension provide light retention, reducing rattling and ensuring smooth re-engagement.
- This design is:
 - Low-cost (only one added plate + small slot)
 - Self-locking and self-releasing
 - Unaffected by actuator timing
 - Fully compatible with the existing link–cylinder mechanism



The End

THANK YOU FOR LISTENING